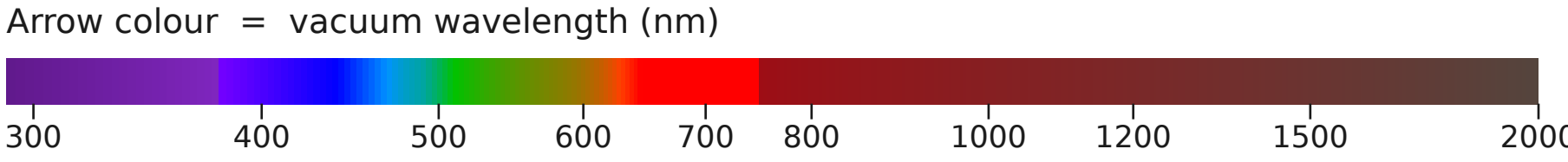
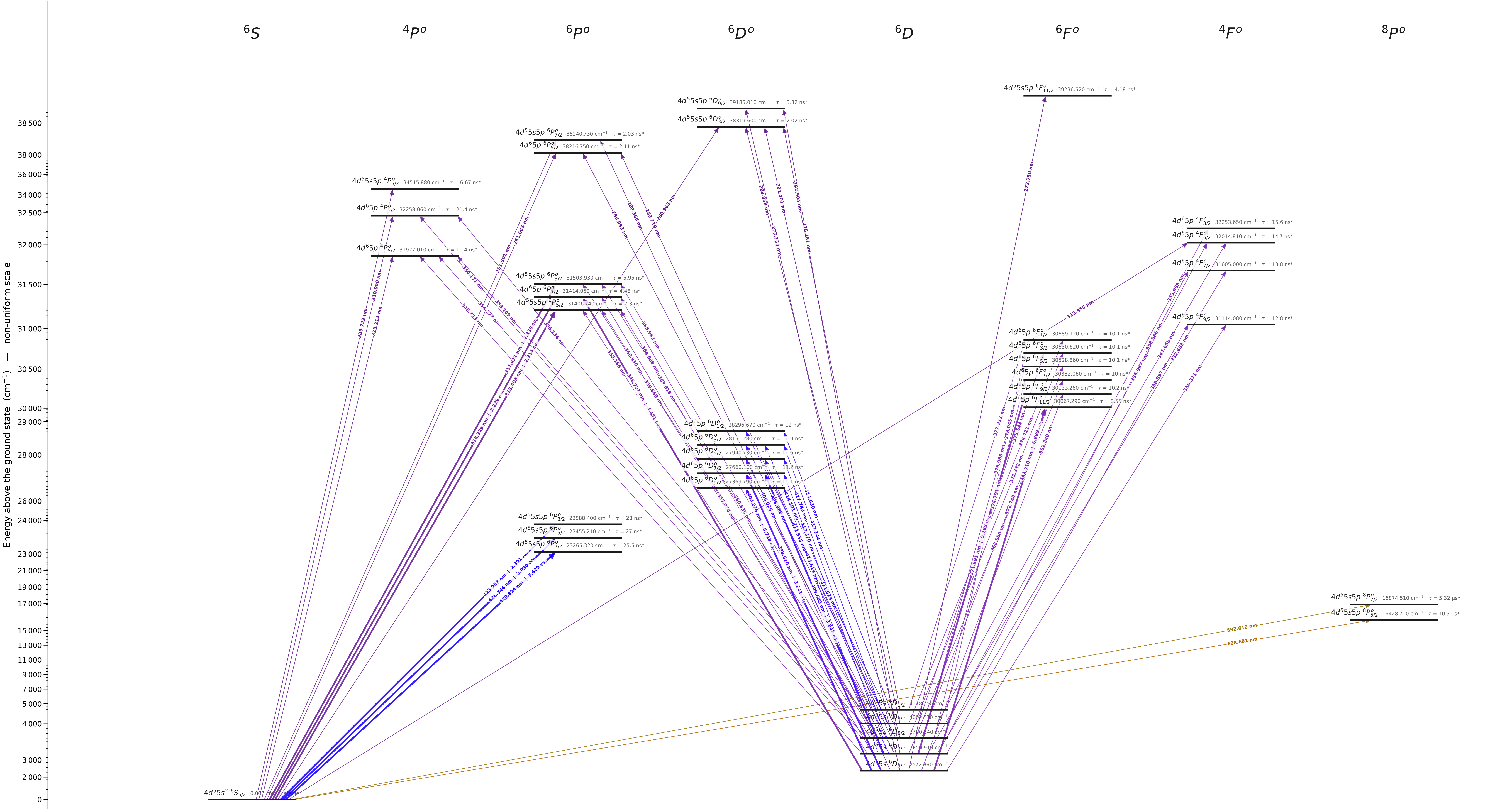


# Technetium

Arrow label: vacuum wavelength | reduced dipole matrix element  $| \langle J || e r || J' \rangle |$  in  $ea_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Energy above the ground state (cm<sup>-1</sup>) — non-uniform scale



## Key transitions

| Transition                                         | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $  \langle J    e r    J' \rangle  $ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                                                          |
|----------------------------------------------------|--------------------|--------------------|-------------|---------------------------------------------------------|----------------------|-------------------------|------------|--------------------------------------------------------------|
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{3/2}^o$ | 423.9372           | 423.8178           | 707.1624    | 2.391                                                   | 3.800e+07            | 5.682                   | 100        | first step of the three-step Ti:sapphire resonance-ioniza... |
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{5/2}^o$ | 426.3445           | 426.2245           | 703.1695    | 3.030                                                   | 4.000e+07            | 5.889                   | 100        | first step of the three-step Ti:sapphire resonance-ioniza... |
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{7/2}^o$ | 429.8243           | 429.7034           | 697.4768    | 3.629                                                   | 4.200e+07            | 6.239                   | 100        | first step of the three-step Ti:sapphire resonance-ioniza... |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

## Isotope shifts

| Transition                                         | $\lambda$ vac (nm) | Isotopes | Shift (MHz) |
|----------------------------------------------------|--------------------|----------|-------------|
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{3/2}^o$ | 423.937            | 97-99    | 308(14)     |
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{5/2}^o$ | 423.937            | 98-99    | 324(14)     |
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{7/2}^o$ | 426.344            | 97-99    | 313(14)     |
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{3/2}^o$ | 426.344            | 98-99    | 327(14)     |
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{5/2}^o$ | 429.824            | 97-99    | 347(14)     |
| $4d^5 5s^2 \ ^6S_{5/2} - 4d^5 5s 5p \ ^6P_{7/2}^o$ | 429.824            | 98-99    | 369(14)     |

Shift =  $\nu$ (first isotope) -  $\nu$ (second isotope).

## <sup>99</sup>Tc hyperfine structure ( $I = 9/2$ )

| Level                      | A (MHz)     | B (MHz) | Shift of each F level from the centre of gravity (MHz)                                                             |
|----------------------------|-------------|---------|--------------------------------------------------------------------------------------------------------------------|
| $4d^5 5s^2 \ ^6S_{5/2}$    | -107.34(8)  | 0.3(20) | F=2: +1476.06 F=3: +1153.95 F=4: +724.51 F=5: +187.76 F=6: -456.26 F=7: -1207.50                                   |
| $4d^5 5s 5p \ ^6P_{3/2}^o$ | 799.99(26)  | -44(4)  | F=1: -15419.93 F=2: -13813.68 F=3: -11405.61 F=4: -8197.28 F=5: -4190.80 F=6: +611.23 F=7: +6205.67 F=8: +12588.87 |
| $4d^5 5s 5p \ ^6P_{5/2}^o$ | 894.08(28)  | 57(4)   | F=2: -12267.29 F=3: -9602.27 F=4: -6042.22 F=5: -1581.38 F=6: +3787.40 F=7: +10072.75                              |
| $4d^5 5s 5p \ ^6P_{7/2}^o$ | 1183.27(27) | -13(3)  | F=3: -9767.75 F=4: -5026.27 F=5: +892.70 F=6: +7983.92                                                             |

Measured A, B constants; sources in the data file.